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基于位置倒排索引的 TDT3 新闻命令行搜索引擎

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1　实验目的

新闻语料规模庞大且持续增长，从中快速找出与某一事件相关的报道，是信息检索最典型的应用场景之一。搜索引擎通过倒排索引把“词到文档”的映射预先组织好，再以相关性打分函数对候选文档排序，使用户只需输入几个关键词，就能在毫秒级时间内得到按相关程度排列的结果。命令行搜索引擎去掉了图形界面和分布式架构，只保留索引、查询解析与排序这几个环节，因此适合用来考察检索系统的基本机制，以及各项设计选择对排序结果的影响。

本实验以 TDT3 语料中的 37,526 篇新闻为对象，从零实现了一个命令行搜索引擎。系统包括文本预处理、位置倒排索引、自由文本与双引号短语混合的查询解析、BM25 与 TF-IDF 两种手写打分函数，以及查询相关摘要生成，能按指定条数输出排名、得分、文档编号和摘要。在五条测试查询上，我们比较了两种打分函数的排序行为，并对词干还原、短语匹配方式、连字符处理和 BM25 长度归一化参数逐一做了对照实验，从实验数据出发解释各项设计对检索结果的影响。

2　实验设置

2.1　数据集

TDT3 语料按日期分为 92 个子目录，共 37,526 个文件，每个文件是一篇 SGML 文档，包含 `DOCNO`、`DOCTYPE`、`TXTTYPE` 和 `TEXT` 四个字段。我们以 `TEXT` 字段作为正文建立索引，以去掉扩展名的文件名（如 `NYT19981231_0439`）作为文档编号。

语料由八个来源的两类文本组成。一类是通讯社稿件，包括纽约时报（NYT，6,871 篇）和美联社（APW，7,338 篇）；另一类是广播节目的字幕或转写稿，包括 CNN（11,954

篇）、ABC、NBC、MSNBC（MNB）、PRI 与美国之音（VOA），每个文件对应节目中的一个片段。按 DOCTYPE 划分，31,276 篇为新闻（NEWS），5,997 篇为节目预告、串场词等杂项（MISCELLANEOUS），253 篇未转写（UNTRANSCRIBED），其中 615 篇正文为空。

两类文本在篇幅上相差悬殊，文档长度因此呈双峰分布，如图 1 所示。全体文档的平均长度为 330.2 个词项，中位数只有 175。左峰在 60 至 80 个词项附近，主要由 CNN 的短片段构成，CNN 文档长度的中位数为 63；右峰在数百个词项处，由通讯社稿件构成，NYT 文档长度的中位数为 837。文档长度跨越两个数量级，而且有大量只有一两句话的短片段，可以预期，不同打分函数对文档长度的归一化方式会显著影响排序结果，这是后文对比 BM25 与 TF-IDF 的主要切入点。此外，广播字幕的大小写和拼写都不规范，预处理阶段需要对词形作统一的规范化。

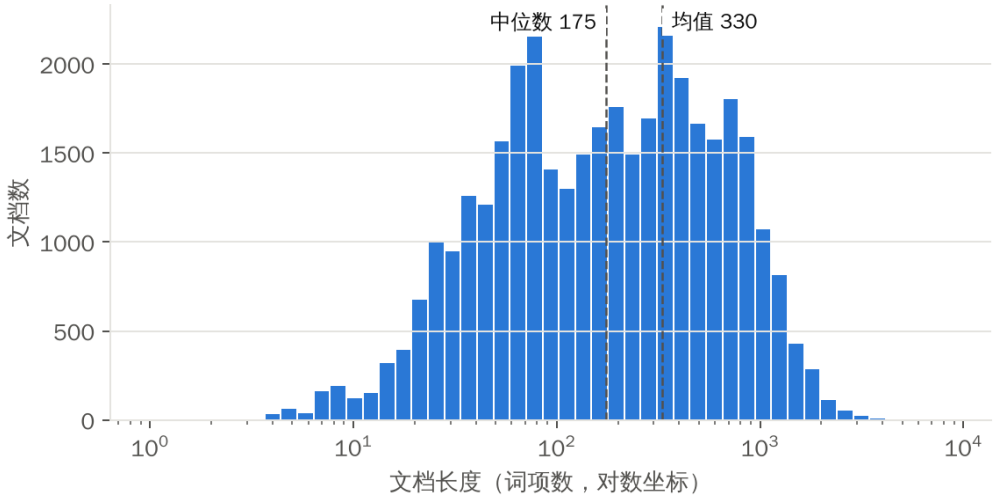


图 1: TDT3 文档长度分布（不含空文档，横轴为对数坐标）

2.2　软件环境

实验所用的软件环境如表 1 所示。

表 1: 实验软件环境

项目	版本
操作系统	Ubuntu 24.04.5 LTS
Python	3.13.18
NLTK	3.10.4
matplotlib	3.11.2
Playwright / Chromium	1.63.0
XeLaTeX	TeX Live 2023

2.3　测试查询与观察指标

测试使用五条查询：Q1 为 `hurricane`，Q2 为 `mitch george`，Q3 为 `bill clinton israel`，Q4 为 `"newt gingrich" down`，Q5 为 `nba strike closed-door`。它们依次覆盖了单词查询、多词自由文本查询、带引号短语的混合查询和带连字符的查询。

由于语料没有针对这些查询的相关性标注，我们改用四个可以直接计算的观察量来比较排序结果：命中文档数，即得分非零的文档数；Top-10 中同时包含全部查询单元（每个查询词以及每个短语）的文档数；Top-10 文档的平均长度；以及两种模型 Top-10 结果的重合篇数。在此基础上，我们逐条阅读 Top-10 文档的正文，对排序的合理性作定性判断。查询延迟取 20 次执行的中位数，包含摘要生成，不含索引加载。

3　方法与实现

3.1　系统组成

系统用 Python 实现，按处理阶段分为七个模块。`porter` 实现 Porter 词干算法；`analyzer` 负责分词与规范化，由索引端和查询端共用；`corpus` 读取 SGML 文档；`index` 构建、保存和加载位置倒排索引；`query` 解析查询；`scoring` 实现 BM25 与 TF-IDF 两种打分函数；`snippet` 生成查询相关摘要；命令行入口 `search.py` 提供一次性查询和交互式查询两种用法。编写的完整内容请参见随附代码。

3.2　文本预处理

预处理的首要原则是让索引端和查询端对同一个词给出同一个词项。围绕这一原则，我们在大小写、数字、标点和词干四个方面分别作出了取舍，实现如清单 1 所示。

清单 1: 分析器的分词正则与规范化逻辑

```
_TOKEN_RE = re.compile(
    r"(?P<num>\d+(?:[.,]\d+)+)"          # 1,000  3.6  1,234.5
    r"|(?P<abbr>(?:[a-z]\.){2,})"         # u.s.  u.n.
    r"|(?P<word>[a-z0-9]+(?:['’ \-][a-z0-9]+)*)"  # words, clinton's, closed-door
)

    def surface_tokens(self, text: str) -> Iterator[Tuple[List[str], int, int]]:
        """Yield (terms, start, end) for every surface token of `text`.

        `terms` has more than one element only for hyphenated words. Offsets
        refer to the original string and are used for snippet highlighting.
        """
        lowered = text.lower()
        for m in _TOKEN_RE.finditer(lowered):
```

```
            kind = m.lastgroup
            tok = m.group(kind)
            if kind == "num":
                if "." in tok and "," in tok:
                    tok = tok.replace(",", "")
                elif "," in tok:
                    tok = tok.replace(",", "")
                yield [tok], m.start(), m.end()
            elif kind == "abbr":
                yield [tok.replace(".", "")], m.start(), m.end()
            else:
                tok = tok.replace("’ ", "'")
                if tok.endswith("'s"):
                    tok = tok[:-2]
                tok = tok.replace("'", "")
                parts = [p for p in tok.split("-") if p]
                if parts:
                    yield [self._normalize_part(p) for p in parts], m.start(),
    ↪ m.end()
```

大小写在这份语料中不承载区分信息，我们把全文统一转为小写。广播字幕的大小写书写相当随意：同一个飓风名，在 VOA 稿件中写作 “Hurricane Georges”，在 NBC 字幕中写作 “hurricane georges”，在 CNN 字幕中又写作 “Hurricane georges”。如果区分大小写，同一实体会被拆成多个词项，而用户输入查询时的大小写同样无从约定。

数字是新闻中有检索价值的信息，我们予以保留，并统一其写法。年份、伤亡人数和金额常常就是用户要找的内容；为了让同一个数值只对应一个词项，分析器删除千位分隔符（“1,000” 变为 `1000`），保留小数点（“3.6” 保持为 `3.6`），并且不对纯数字做词干还原。

标点本身不携带检索信息，我们将其删除，但单独处理了三类会破坏词项一致性的情形。由点号连接的缩写合并为一个词（“U.S.” 变为 `us`）；英文所有格 “’s” 被去掉，其余撇号直接删除（“Clinton’s” 变为 `clinton`，“don’t” 变为 `dont`）；带连字符的词拆成若干部分，按顺序占据相邻位置（“closed-door” 变为位置相邻的 `closed` 与 `door`）。经过连字符拆分，原文中的 “closed door” 与 “closed-door” 得到相同的词项序列，可以被同一个短语查询命中。

词形变化会把同一概念分散到多个词项上，我们用自行实现的 Porter 算法做词干还原，使 “hurricane” 与 “hurricanes”、“strike” 与 “strikes”、“closed” 与 “closing” 归并为同一词项。以 NLTK 中遵循原论文的 `ORIGINAL_ALGORITHM` 模式为参照，我们对不做词干还原的索引中的 110,617 个纯字母词逐一比对，两者只在 27 个两字母词（如 `us`、`is`）上不同，原因是我们沿用了 Porter 参考实现 “长度不超过 2 的词不做处理” 的约定。

停用词予以保留。常见词的权重已经由逆文档频率压低：`down` 出现在 6,941 篇文

档中，BM25 逆文档频率只有 1.69，而 `nba` 为 4.89。另一方面，`down` 恰好是 Q4 的查询词，删除停用词会使 Q4 退化为单纯的短语查询。

3.3　位置倒排索引

短语查询要求索引记录词项的出现位置，因此我们构建了位置倒排索引。每个词项对应三个并行的无符号整数数组：`docs` 按升序记录包含该词项的文档编号，`tfs` 记录该词项在各文档中的出现次数，`pos` 按文档顺序依次存放全部出现位置，第 k 篇文档的位置区间由 `tfs` 的前缀和确定。以紧凑数组代替嵌套列表存放六百万条倒排记录，降低了序列化和加载的开销：索引文件为 101.7 MiB，加载一次只需 0.4 至 0.5 秒。构建过程如清单 2 所示，同一趟扫描中还为每篇文档预先计算了打分所需的文档长度 $|d|$ 和文档向量长度 $\|d\|$。

清单 2: 索引构建的主循环

```
for n, path in enumerate(iter_files(data_root)):
    docid, doctype, text = read_doc(path)
    d = len(idx.doc_names)
    idx.doc_names.append(docid)
    idx.doc_paths.append(os.path.relpath(path, data_root))
    idx.doc_types.append(doctype)

    local: Dict[str, List[int]] = defaultdict(list)
    for p, term in enumerate(analyzer.terms(text)):
        local[term].append(p)
    length = sum(len(v) for v in local.values())
    norm_sq = 0.0
    for term, plist in local.items():
        pst = postings[term]
        pst.docs.append(d)
        pst.tfs.append(len(plist))
        pst.pos.extend(plist)
        w = 1.0 + math.log10(len(plist))
        norm_sq += w * w
    idx.doc_len.append(length)
    idx.doc_norm.append(math.sqrt(norm_sq))
```

3.4　查询解析与短语软匹配

查询解析器先取出所有成对双引号括起的短语，再把剩余文本作为自由文本处理，结果是一组“查询单元”，每个单元是一个词项或一个短语，实现如清单 3 所示。

清单 3: 查询解析：短语与连字符词的处理

```
def add_group(terms, hyphenated=False):
    if not terms:
        return
    if len(terms) == 1:
        q.terms.update(terms)
        return
    mode = hyphen_mode if hyphenated else "phrase+parts"
    if mode in ("phrase+parts", "parts"):
        q.terms.update(terms)
    if mode in ("phrase", "phrase+parts"):
        q.phrases.append(tuple(terms))
        q.phrase_terms.update(terms)

# quoted phrases first, then whatever is left is free text
rest = []
last = 0
for m in _PHRASE_RE.finditer(text):
    rest.append(text[last:m.start()])
    add_group(analyzer.terms(m.group(1)))
    last = m.end()
rest.append(text[last:])
free = re.sub(r"[\" “” ]", " ", " ".join(rest))  # unbalanced quotes
for terms, _, _ in analyzer.surface_tokens(free):
    add_group(terms, hyphenated=True)
return q
```

引号短语采用软匹配：短语作为一个单元参与打分，短语中的每个词也作为独立单元加入查询。Q4 因此被解析为短语单元 `(newt, gingrich)` 以及词项单元 `newt`、`gingrich` 和 `down`。含有完整短语的文档额外获得短语单元的得分，只零散含有这些词的文档仍能被检索到，但排名靠后。要求文档必须含有短语的硬匹配保留为 `--strict-phrase` 选项，用于对照实验。短语单元的文档频率和文档内频率通过位置求交得到：先从文档频率最小的词出发求出同时含有全部词的候选文档，再在每篇候选文档中检查位置 $p, p+1, \ldots$ 是否依次出现短语的各个词。

连字符词默认只构成一个短语单元。如果沿用引号短语的规则，一个连字符词会在查询中占据三个单元：Q5 的 `closed-door` 由短语、`close` 和 `door` 共同贡献 11.19 的 BM25 逆文档频率（分别为 5.62、2.06 和 3.51），是 `nba` 的 4.89 的两倍多，排在前列的几乎全是联合国安理会“闭门磋商”的报道。连字符词在语义上只是查询中的一个词，因此我们让它只占一个单元；原规则以及“只拆成独立词项”的规则通过 `--hyphen` 选项保留，对照结果见第 4.5 节。

3.5　相关性打分函数

两种打分函数都按查询单元逐个累加得分：对每个单元遍历其倒排表，把它对各文档的贡献累加到以文档编号为键的累加器中，最后用堆选出得分最高的若干篇文档。下文以 N 表示文档总数，df_u 表示含有单元 u 的文档数，$\mathrm{tf}_{u,d}$ 表示 u 在文档 d 中的出现次数，qtf_u 表示 u 在查询中的出现次数。

BM25 的打分公式为

$$\mathrm{score}(d,q)=\sum_{u\in q}\mathrm{qtf}_u\cdot\mathrm{IDF}(u)\cdot\frac{\mathrm{tf}_{u,d}\,(k_1+1)}{\mathrm{tf}_{u,d}+k_1\left(1-b+b\,\dfrac{|d|}{\mathrm{avgdl}}\right)},\qquad(1)$$

$$\mathrm{IDF}(u)=\ln\!\left(1+\frac{N-\mathrm{df}_u+0.5}{\mathrm{df}_u+0.5}\right),$$

其中 $|d|$ 为文档长度，avgdl 为平均文档长度，k_1 控制词频饱和的速度，b 控制长度归一化的强度，我们取 $k_1=1.5$、$b=0.75$。当 $\mathrm{tf}_{u,d}$ 趋于无穷时，单个单元的贡献趋于上限 $\mathrm{IDF}(u)\,(k_1+1)$；对 Q1 的 `hurricane`，这一上限为 $4.022\times 2.5=10.06$。实现如清单 4 所示。

清单 4: BM25 打分

```
def score(self, index, query, restrict=None):
    N = index.num_docs
    K = self._length_norm(index)
    k1p1 = self.k1 + 1
    acc = defaultdict(float)
    for unit, qw, docs, tfs in _units(index, query, self.phrase_weight):
        df = len(docs)
        w = qw * math.log(1 + (N - df + 0.5) / (df + 0.5))
        for d, tf in zip(docs, tfs):
            if restrict is None or d in restrict:
                acc[d] += w * tf * k1p1 / (tf + K[d])
    return acc
```

TF-IDF 向量空间模型采用 SMART 记法中的 lnc.ltc 方案。查询向量中单元 u 的权重为 $w_{q,u}=(1+\log_{10}\mathrm{qtf}_u)\cdot\log_{10}(N/\mathrm{df}_u)$，经余弦归一化后与文档向量做内积；文档向量的权重为对数词频除以文档向量长度：

$$w_{d,u}=\frac{1+\log_{10}\mathrm{tf}_{u,d}}{\|d\|},\qquad \|d\|=\sqrt{\sum_{t\in d}(1+\log_{10}\mathrm{tf}_{t,d})^2},\qquad \mathrm{score}(d,q)=\sum_{u\in q}\frac{w_{q,u}}{\|q\|}\,w_{d,u}.\qquad(2)$$

实现如清单 5 所示。

清单 5: TF-IDF（lnc.ltc）打分

```
def score(self, index, query, restrict=None):
    N = index.num_docs
    norms = index.doc_norm
    units = list(_units(index, query, self.phrase_weight))
    wq = []
    for unit, qw, docs, tfs in units:
        idf = math.log10(N / len(docs))
        wq.append((1 + math.log10(qw)) * idf if qw >= 1 else qw * idf)
    qnorm = math.sqrt(sum(w * w for w in wq)) or 1.0
    acc = defaultdict(float)
    for (unit, qw, docs, tfs), w in zip(units, wq):
        w /= qnorm
        if w == 0.0:
            continue
        for d, tf in zip(docs, tfs):
            if restrict is None or d in restrict:
                acc[d] += w * (1 + math.log10(tf)) / norms[d]
    return acc
```

3.6　摘要生成与命令行交互

摘要应当展示文档中与查询最相关的部分，因此我们为每篇结果挑选一个包含 36 个连续词的窗口：优先覆盖最多的不同查询词，其次包含完整短语，再次命中次数最多，均相同时取最靠前的窗口。终端输出中，命中的查询词以黄色粗体高亮，文档编号以青色显示。

命令行支持一次性查询和交互式查询两种用法。一次性查询直接在命令后写出查询词，例如 `./search --hits=27 new york san francisco`，`--hits` 指定返回条数，`--model` 在 `bm25` 与 `tfidf` 之间切换。交互模式只加载一次索引，随后可以在提示符“`#`”后连续输入 `search --hits=10 "newt gingrich" down` 这样的命令，双引号短语可以直接书写；每条查询依次编号，并以“`q1 = 查询 (took 秒数 seconds)`”作为结果的标题行。

4　实验结果

4.1　索引构建

表 2 列出了两套索引的规模：一套启用 Porter 词干还原，为系统默认配置；另一套不做词干还原，用于消融对照。

表 2: 索引规模与构建开销

指标	启用词干还原	不做词干还原
文档数	37,526	37,526
词项总数（全部位置）	12,389,223	12,389,223
词表大小	92,128	122,340
只出现一次的词项数	34,598	43,006
倒排记录数（词项–文档对）	6,006,714	6,361,761
索引文件大小	101.7 MiB	106.4 MiB
构建时间	约 22 秒	约 22 秒

词干还原对词表的压缩远大于对倒排记录的压缩：词表从 122,340 个词项减少到 92,128 个，降幅 24.7%，倒排记录数只减少了 5.6%。被归并的大多是同一个词的屈折变形，如 “hurricane” 与 “hurricanes”，它们常常同时出现在同一篇文档中，归并后在这篇文档里只留下一条倒排记录，因此记录数的降幅远小于词表。位置信息的总量与是否做词干还原无关，两套索引都保存了 1,238 万个位置，它们构成了索引文件的主体，两套索引的文件大小也因此只相差 4.7 MiB。

4.2 五条测试查询的 BM25 检索结果

图 2 至图 6 是默认配置（BM25，$k_1 = 1.5$，$b = 0.75$，启用词干还原）下五条测试查询的 Top-10 控制台截图，表 3 汇总了两种打分函数下的观察量。

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 hurricane
q1 = hurricane (took 0.013 seconds)
[bm25, 672 matching documents]

01 [8.8695] CNN19981019_1600_0586
...the city of mazatlan. Mexico's government has sent 2,000 soldiers and
shipments of aid supplies to the area. Meanwhile, hurricane Lester's winds fell
to about 90 miles per hour, and a Previous hurricane warning was downgraded to a
hurricane ...

02 [8.8151] CNN19981008_1130_0447
Hurricane georges is now the third costliest hurricane on record. Actuaries
estimate georges caused $2.5 billion in insured losses throughout the U.S. and
its island territories. Puerto Rico was hardest hit with $1.7 billion worth of
...

03 [8.7823] ABC19981130_1830_1182
This is the end of the hurricane season, according to the calendar, anyway. It
was the deadliest in 200 years and one of the most enduring. There is still
hurricane Nicole, way out in the atlantic, the tenth hurricane ...

04 [8.7711] VOA19981201_0500_0794
...year. But it is also because the Federal Hurricane Research Agencies had
better equipment including more so-called hurricane hunter airplanes to monitor
the storms. But the storm watchers say they need to rethink the way they measure
storms. Hurricanes ...

05 [8.7589] CNN19981018_1000_0421
Two hurricanes are threatening Mexico's pacific coast. Hurricane Lester has made
an abrupt left turn overnight and is heading westward, away from the southern
part of the coast. But forecasters warn it could make a turn ...

06 [8.7174] CNN19981018_1300_0286
Hurricane warnings are posted along Mexico's pacific coast. Forecasters say twin
hurricanes are lurking. At last report, hurricane Lester was about 195 miles
southeast of acapulco. Its maximum sustained winds are about 100 miles an hour
...

07 [8.7153] NBC19981027_1830_0212
...The highest winds ever recorded in a hurricane hitting the U.S. 195 miles an
hour. In killer hurricane camille, in 1969. Slamming into the Mississippi coast
all the way into Virginia. At the national center for atmospheric research,
hurricane ...

08 [8.6806] ABC19981027_1830_0173
It is as we said, very unusual for a hurricane to reach the level of category 5.
hurricane Mitch is already being blamed for the deaths of at least ten people.
Here's ABC's Bill Blakemore. It's ...

09 [8.6726] MNB19981130_2100_1560
...what turns out to have been one of the busiest hurricane soons ever. It was
$DEADLIEST season since 178 O'when people were killed in the caribbean. Almost
all related deaths were related to hurricane Mitch. There were 10 hurricanes ...

10 [8.6599] CNN19981130_1600_0465
One of the deadliest hurricane seasons ever ends today. 14 tropical storms
formed in the atlantic basin between June 1st and today. Ten grew to hurricane
strength. The two most destructive storms were Mitch and georges ...
```

图 2: Q1 `hurricane` 的 BM25 Top-10 结果

Q1 命中 672 篇文档, Top-10 全部是飓风报道, 涉及 Lester、Georges、Mitch、Nicole 等飓风和 1998 年飓风季的总结。单词查询的排序主要由文档长度决定：十篇文档的得分集中在 8.66 至 8.87 之间，都已逼近得分上限 10.06，词频的作用基本饱和。排名第

1 的文档长 86 个词项，“hurricane” 出现 5 次；排名第 4 的 VOA 报道长 577 个词项，出现 16 次，靠更高的词频抵消了长度惩罚。

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 mitch george
q1 = mitch george (took 0.013 seconds)
[bm25, 1861 matching documents]

01 [13.6995] NBC19981105_1830_0070
...Mexico for almost two weeks now, it leaves death everywhere. Today, Mitch
plowed across south Florida with surprising strength -- eight inches of rain
70-mile-an-hour winds battered the keys, which still haven't recovered from
hurricane georges six weeks ago. Mitch ...

02 [13.1973] CNN19981105_0130_0429
...gulf and atlantic sides of Florida. Mitch's 45-mile-an-hour winds caused
tornadoes that touched down in key largo, and one that took the roof off a motel
in islamorada. The keys are still recovering from damage inflicted by hurricane
georges ...

03 [12.8096] VOA19981026_1600_0283
...the storm's 250 kilometer per hour winds make Mitch far more powerful than
Hurricane Georges, which killed more than 500 people last month. James Lewis
Free of the National Hurricane Center says Mitch is a very powerful storm. Mitch
...

04 [12.2860] VOA19981109_0500_0302
International aide shipments are pouring into Central America as workers
continue to sort through devastation caused by Hurricane Mitch. The American Red
Cross shipped enough food Sunday to feed 1000 families for a month in the ...

05 [12.1688] VOA19981123_0500_2570
...Hilary Rodham Clinton has promised more U.S. aid to Haiti to help repair
damage caused by Hurricane George. Ms. Clinton announced Sunday the United
States will add $2.1 million to an existing $12 million storm relief package.
Hurricane George ...

06 [12.1546] CNN19981130_2130_0482
One of the most destructive hurricane seasons ever ends today. 14 tropical
storms formed in the atlantic basin between June 1st and today. Ten grew to
hurricane strength. The two most destructive storms were Mitch and georges ...

07 [11.7596] CNN19981201_0130_0318
...the atlantic basin between June 1st and november 30th. Ten grew to hurricane
strength. The two most destructive storms were Mitch and georges. They caused
more than 11,000 deaths and did $2.5 billion in property damage. Forecasters say
Mitch ...

08 [11.5517] APW19981025_0908
...some streets in the Jamaican capital Sunday. Under dark, menacing skies,
seaports closed and buses and taxis stayed off Kingston's streets as Mitch, the
13th named storm of the Atlantic hurricane season, swirled in the western
Caribbean Sea. Mitch ...

09 [11.4604] NBC19981224_1830_1277
...remembering it could be a lot worse. Almost two months ago, hurricane Mitch
devastated much of central America. NBC's George Lewis is there. Among the two
and a half million people in central America made homeless by hurricane Mitch
...

10 [11.4493] CNN19981130_1600_0465
One of the deadliest hurricane seasons ever ends today. 14 tropical storms
formed in the atlantic basin between June 1st and today. Ten grew to hurricane
strength. The two most destructive storms were Mitch and georges ...
```

图 3: Q2 `mitch george` 的 BM25 Top-10 结果

Q2 的结果集中在同时提及飓风 Mitch 与飓风 Georges 的报道上，十篇文档都同时含有 `mitch` 与 `georg` 两个词项。词干还原在这里起了关键作用：“georges” 与查询词 “george” 被还原为同一个词项 `georg`，查询由此覆盖了飓风 Georges 的名字。排名第 5 的 VOA 文档把飓风名拼写为 “Hurricane George”，同样被召回；排名第 9 的 NBC 文档在报道飓风 Mitch 时署名记者为 “George Lewis”，属于名字上的巧合。

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 bill clinton israel
q1 = bill clinton israel (took 0.020 seconds)
[bm25, 7474 matching documents]

01 [14.0375] VOA19981208_0500_0050
I'm Bill Hazard, VOA news. U.S. mediator Dennis Ross is beginning a mission to
the mideast aimed at diffusing tensions between Israel and the Palestinians
ahead of President Clinton's trip to the region. Israel has suspended ...

02 [13.8595] APW19981212_0565
Palestinian police helped quell anti-Israel protests on Saturday, hours before
U.S. President Bill Clinton was to arrive in the region _ a sign of Palestinian
leader Yasser Arafat's determination to smooth the visit. In
Palestinian-administered Bethlehem, Palestinian ...

03 [13.5149] PRI19981214_2000_0055
...this is Pamela McCall in the BBC Newsroom in London. The Palestinian's most
senior ruling body has once again renounced clauses calling for the destruction
of Israel. The Palestine National Council took the vote in front of Bill Clinton
...

04 [13.4870] APW19981213_0389
The speaker of the Iranian Parliament said Sunday that U.S. President Bill
Clinton's planned visit to Gaza was part of a ``dangerous conspiracy,'' the
official news agency reported. Clinton, who arrived in Israel late Saturday,
plans ...

05 [12.9325] CNN19981017_1000_0047
...an additional 13% of the West Bank. Israel controls the routes to many
Palestinian-controlled areas which, experts say, has been a problem for
autonomy-minded Palestinians. Since today is the Jewish sabbath, today's
negotiation sessions in Maryland are being billed ...

06 [12.8687] APW19981217_0276
...from Pennsylvania, did not disclose the contents of Israeli Prime Minister
Benjamin Netanyahu's letter to Assad. Specter, who had accompanied President
Bill Clinton during his visit to Israel and the Palestinian territories earlier
this week, has shuttled between Israel ...

07 [12.6765] APW19981120_1198
...over that area. _Mid-December: President Bill Clinton attends a meeting of
the Palestinian National Council in Gaza, where the body is to reaffirm the
revocation of clauses in the PLO charter calling for Israel's destruction. _End
of January: Israel ...

08 [12.6024] APW19981018_0231
JERUSALEM (AP) _ U.S. President Bill Clinton aims to put pressure on Israel by
threatening to recognize a Palestinian state this spring if the two sides don't
reach an agreement at the Mideast summit outside Washington, Israel's ...

09 [12.5728] APW19981101_0848
JERUSALEM (AP) _ U.S. President Bill Clinton said in an interview Sunday that
the negotiations that led to the Israeli-Palestinian peace agreement last month
were volatile and heated at times. In an interview with Israel's Channel 2
television, Clinton ...

10 [12.5207] APW19981212_0833
...jostles the Palestinian red, white, black and green. Atop Jerusalem telephone
poles, the Stars and Stripes gently lap Israel's Star of David. Roads were ready
Saturday for the presidential motorcade in Israel and the Palestinian areas _
but Bill Clinton ...
```

图 4: Q3 `bill clinton israel` 的 BM25 Top-10 结果

Q3 命中 7,474 篇文档。Top-10 全部同时含有三个查询词，其中七篇来自美联社，内容集中在 1998 年克林顿访问以色列和加沙、推动中东和谈等事件上。结果中也出现

了一词多义和词干还原造成的匹配偏差：排名第 1 的 VOA 文档以 “I’m Bill Hazard, VOA news” 开头，“Bill” 是记者的名字；排名第 5 的 CNN 文档命中的是 “being billed”，经词干还原后与 `bill` 相同。这两篇文档的主体仍是以色列与巴勒斯坦的谈判。

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 "newt gingrich" down
q1 = "newt gingrich" down (took 0.020 seconds)
[bm25, 7212 matching documents]

01 [29.1755] CNN19981107_1300_0109
...and the White House. That's how one republican describes the decision by Newt
Gingrich to step down as speaker of the house. He also plans to give up his
congressional seat as well. In a statement released yesterday, Gingrich ...

02 [27.4158] CNN19981110_0130_0049
...leader. Army says he's way ahead, but ex-wide receiver Steve Largent isn't
ready to take a pass. And Jennifer Dunn just got into the game. We have such a
great opportunity now with Newt Gingrich's deciding to step down ...

03 [26.8852] CNN19981110_1130_0215
...him off. things are beginning to settle down a little, but for a while there
in the last few days, one of the few house republicans who was not considering a
run for a leadership post was newt gingrich ...

04 [26.8445] NBC19981107_1830_0379
...Clinton. I applaud Newt for saying I am not going to hold back my party, I am
not going to hold back the people of this country like our president is doing.
Gingrich's supporters say he may be down ...

05 [26.6286] NBC19981106_1830_0071
...as George bush called it down in Texas. The republicans in the house believe
there was a valuable lesson learned there, that Newt Gingrich had to go, they
had to start fresh. One of his closest friends said Newt ...

06 [26.4226] CNN19981107_1000_0932
A check of our top stories -- house speaker Newt Gingrich says he plans to
resign from Congress in the wake of the disappointing results for republicans
from this week's elections. Sources say Gingrich will step down ...

07 [26.2170] CNN19981105_1130_0337
Meanwhile, house speaker Newt Gingrich is taking some heat for the GOP election
strategy. Republicans warned Gingrich yesterday of a "dump Newt" movement among
house members. Gingrich is downplaying talk of any challenge

08 [25.4900] NBC19981105_1830_0603
...results of election has thrown republicans into turmoil and put the job of
speaker Newt Gingrich on the line tonight. More from Gwen Ifill. Secluded from
public view at his suburban Atlanta home, sources say house speaker Newt
Gingrich ...

09 [25.3649] CNN19981107_1600_0933
Checking the top stories -- house speaker Newt Gingrich confirmed he is stepping
down from his post. He is also resigning from Congress because he does not want
to be a Distraction for the new leaders

10 [25.3072] CNN19981106_2130_0930
Checking the top stories -- sources say house speaker Newt Gingrich is resigning
from Congress. They say he plans to step down as speaker of the house and resign
his house seat before Congress convenes in January
```

图 5: Q4 `"newt gingrich" down` 的 BM25 Top-10 结果

Q4 的十篇结果全部含有完整短语 “newt gingrich”，其中九篇同时含有 `down`，内容集中在 1998 年 11 月金里奇宣布辞去众议院议长（step down）的报道上。第 1 名得

分 29.18，明显高于 Q1 至 Q3 的最高分，这是短语单元与两个组成词各自贡献得分的结果，体现了软匹配对含完整短语文档的加权。

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 nba strike closed-door
q1 = nba strike closed-door (took 0.024 seconds)
[bm25, 2716 matching documents]

01 [12.4102] CNN19981014_1130_0513
...other stuff that I probably wouldn't see with the basketball. It took major
league baseball several years to get its fans back after the players went on
strike. It'll be interesting to see how long it takes the NBA ...

02 [12.2479] VOA19981222_0500_0119
Meanwhile, a divided U.N. Security Council resumes closed-door talks today on
how to go forward in eliminating Iraq's banned weapons of mass destruction. The
talks follow last week's U.S.-led air strikes against Iraq triggered by ...

03 [12.1693] NBC19981017_1830_1375
...The average hockey salary is $1.2 million. And ticket prices have gone up. A
world series ticket is up to $150. A courtside seat for the NBA championship in
Chicago was $475. Despite the huge contracts in the NBA ...

04 [12.0955] CNN19981209_0130_0754
Top stories are next. And later, in consumer news, another basketball tradition
falls by the wayside in the wake of the NBA players strike. For the Energizer
doublebarrel flashlight-- unique double barrel design that extends battery ...

05 [11.4955] CNN19981209_0130_0984
Many basketball fans have been disappointed by the lingering players' strike.
Now, the NBA is apologizing for another cancellation. Tony Guida has the story.
The NBA wrote another embarrassing chapter in its history tuesday, canceling its
...

06 [11.2165] MNB19981124_2100_3227
...has been laying off workers in a business that's been slipping for two years.
The NBA license division has been soft for a while now and retailers have been
cautious because of that. The optimism or expectation that NBA ...

07 [11.1339] NYT19981021_0024
...generated this year by Mark McGwire and Sammy Sosa's home run race and the
Yankees' 114 regular-season victories, baseball's attendance figures are still
lower than they were before the strike. Still, with the resurgence baseball has
enjoyed, the NBA's ...

08 [10.9627] CNN19981029_1600_1327
194 NBA games have been lost to labor strike and the team owners and players
union still have not reached agreement on a new contract. Both sides planned to
talk later today by phone after Wednesday's ...

09 [10.7569] APW19981104_1127
...France is jokingly known, was in its third week of a strike. Library
employees, angry over computer glitches at the high-tech facility that opened to
researchers less than a month ago, are demanding improved working conditions and
closed doors ...

10 [10.6397] NYT19981211_0412
...belated start to the season was unclear, but both parties have apparently
agreed to stop the public posturing and make their bargaining sessions private.
NBA deputy commissioner Russ Granik said the two sides had pledged secrecy
behind closed doors ...
```

图 6: Q5 `nba strike closed-door` 的 BM25 Top-10 结果

Q5 的 Top-10 中有八篇与 NBA 停摆有关，另外两篇是联合国安理会的闭门磋商（第 2 名）和埃菲尔铁塔员工罢工（第 9 名）。三个查询单元在语料中几乎不共现：同时

含有 `nba` 和短语 `closed door` 的文档只有 2 篇，同时含有全部三个单元的只有第 10 名这 1 篇 NYT 报道，原文写道 NBA 劳资双方 “pledged secrecy behind closed doors”。Top-10 中 “全部单元命中” 的文档数为 1，已经达到语料所能提供的上限。

表 3: 五条测试查询的观察量（Top-10，延迟为 20 次执行的中位数）

		BM25				TF-IDF（lnc.ltc）				
查询	命中数	全命中	平均长度	延迟	无摘要	全命中	平均长度	延迟	无摘要	重合
Q1	672	10	180.5	3.8	0.15	10	29.4	0.7	0.16	1
Q2	1,861	10	168.9	4.0	0.44	3	37.3	1.2	0.44	3
Q3	7,474	10	277.1	8.1	2.01	3	64.0	3.4	2.02	3
Q4	7,212	9	262.1	8.5	2.83	0	23.7	3.7	2.97	1
Q5	2,716	1	389.6	13.8	2.66	0	17.8	3.2	2.64	0

“全命中” 为 Top-10 中包含全部查询单元的文档数；“平均长度” 以词项计，语料平均为 330.2；延迟单位为毫秒，“无摘要” 为不生成摘要时的延迟；“重合” 为两种模型 Top-10 的共同文档数。

查询耗时主要花在摘要生成上。五条查询的完整延迟都在 14 毫秒以内，去掉摘要生成后降到 0.15 至 3 毫秒，打分本身的开销很小。

4.3 BM25 与 TF-IDF 的排序对比

两种打分函数的排序差异很大，而且差异集中体现在文档长度上。五条查询的 Top-10 平均只有 1.6 篇重合，Q5 一篇也不重合。图 7 把两种模型返回的每篇文档的长度画在同一坐标中：BM25 的 Top-10 平均长度在 169 至 390 个词项之间，接近语料平均值 330.2；TF-IDF 的 Top-10 平均长度只有 18 至 64 个词项，五条查询的 Top-10 文档没有一篇超过 150 个词项。TF-IDF 的五组 Top-10 截图见附录 A。

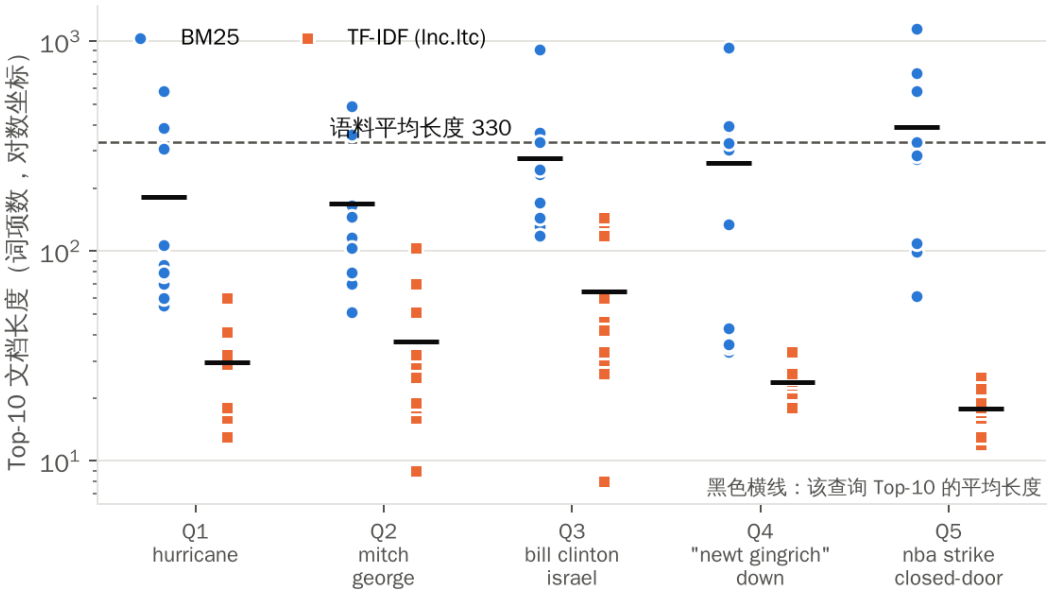


图 7: 两种打分函数 Top-10 文档的长度分布

TF-IDF 的 Top-10 几乎都是广播节目中的一两句话。Q4 的十篇结果全部来自 CNN，都是 “With Newt Gingrich set to leave Congress, ...” 这类 18 至 33 个词项的节目预告，没有一篇含有 `down`；Q5 的十篇结果中有九篇来自 CNN，都是 NBA 停摆的简讯，但没有一篇提到罢工或闭门会议。更极端的例子出现在 Q1：TF-IDF 排名第 1 的是一句 13 个词项的体育预告 “the hurricanes in new york, facing the islanders”，指的是冰球队 Carolina Hurricanes 的比赛。Q3 的第 1 名 “When we continue – peace and politics in Israel” 只有 8 个词项，既不含 `bill` 也不含 `clinton`。

4.4　BM25 长度归一化参数的影响

为确认 BM25 结果的长度特征来自参数 b，我们固定 $k_1 = 1.5$，令 b 在 0 到 1 之间取五个值，记录各查询 Top-10 的平均文档长度，结果如图 8 和表 4 所示。

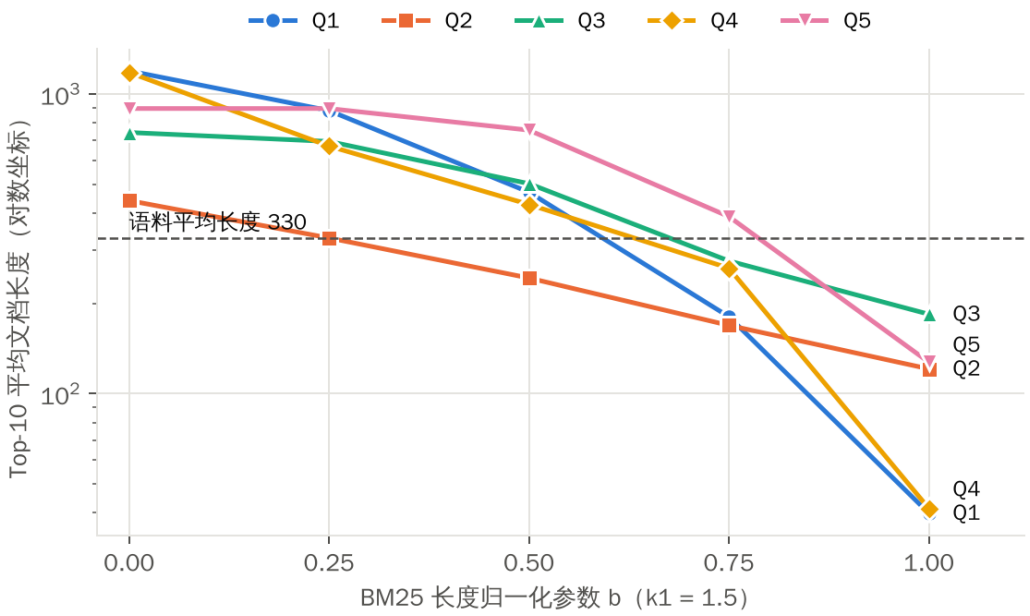


图 8: BM25 参数 b 对 Top-10 平均文档长度的影响

表 4: 不同 b 取值下 Top-10 的平均文档长度（词项数）

b	Q1	Q2	Q3	Q4	Q5
0.00	1193.0	440.5	743.3	1178.8	893.5
0.25	879.5	330.5	693.8	669.4	893.5
0.50	469.9	243.0	501.7	427.5	756.6
0.75	180.5	168.9	277.1	262.1	389.6
1.00	39.9	120.9	184.0	41.4	127.3

b 直接决定了 BM25 偏好的文档长度。b 从 0 增大到 1 时，五条查询 Top-10 的平均长度都单调下降，跨度达一个数量级。$b = 0$ 时 BM25 完全不考虑文档长度，Top-10 被上千词项的长文占据；$b = 1$ 时，Q1 和 Q4 的 Top-10 平均长度降到约 40 个词项，已接近 TF-IDF 的水平。我们采用的 $b = 0.75$ 使 Top-10 平均长度落在 169 至 390 之间，处于语料平均长度附近。

4.5　预处理与查询设计的对照实验

词干还原。词干还原提高了全部五条查询的召回，对排序影响最大的是 Q2。表 5 比较了启用与不启用词干还原时 BM25 的命中文档数和 Top-10 重合篇数。

表 5: 词干还原对 BM25 检索的影响

查询	被归并的主要变形（不还原时的文档频率）	还原后词项（文档频率）	命中（还原）	命中（不还原）	Top-10 重合
Q1	hurricane 621，hurricanes 106	hurrican 672	672	621	6
Q2	george 1,225，georges 123	georg 1,353	1,861	1,752	3
Q3	bill 2,348，bills 567，billed 85	bill 2,849	7,474	7,043	9
Q4	down 6,830，downs 111，downed 70	down 6,941	7,212	7,102	10
Q5	strike 1,097，strikes 1,275	strike 2,345	2,716	1,418	7

Q5 的命中数几乎翻倍，因为“strikes”的文档频率甚至高于“strike”。Q2 不做词干还原时，“george”无法匹配“georges”，Top-10 只有 3 篇与默认配置相同，且没有一篇含有“georges”：其中 5 篇因 NBC 记者 George Lewis 的署名而命中 `george`，2 篇提到的是前总统 George Bush，只有 1 篇是把飓风名拼写为“Hurricane George”的 VOA 报道。

短语的软匹配与硬匹配。在 Q4 上，两种方式给出的 Top-10 完全相同，BM25 和 TF-IDF 都是如此。硬匹配只保留含有完整短语的 406 篇文档，软匹配则对 7,212 篇文档打分，但软匹配下 Top-10 的十篇文档本来就全部含有该短语。全语料中同时含有 `newt` 和 `gingrich` 的文档共 409 篇，其中只有 3 篇二者不相邻。

连字符词的匹配方式。连字符词的处理规则对 Q5 的排序影响很大，表 6 比较了三种规则。

表 6: Q5 中连字符词的三种处理规则（Top-10 中的文档数）

打分函数	连字符词规则	含 nba 的文档	含短语 closed door 的文档
BM25	短语 + 组成词（与引号短语相同）	1	9
	仅短语（默认）	8	3
	仅组成词	8	2
TF-IDF	短语 + 组成词（与引号短语相同）	4	6
	仅短语（默认）	10	0
	仅组成词	10	0

在 BM25 下，把连字符词同时展开为短语和组成词时，Top-10 中只有 1 篇与 NBA 有关，含有“closed door”的文档占了 9 篇；改为只构成一个短语单元后，NBA 相关文档增至 8 篇，同时仍有 3 篇含有“closed door”，其中包括唯一同时命中三个查询单元的 NYT 报道。只拆成独立组成词时，NBA 相关文档同样为 8 篇，含有“closed door”的文档降为 2 篇。这一规则的第 1 名是一篇高度相关的 CNN 报道，原文为“NBA owners and players have closed the door to the media by conducting secret meetings”，“closed”与“door”之间隔着“the”，在仅短语规则下只排在第 119 名；但它的第 2、4、7、10 名得分，是因为词干还原后的 `close` 匹配了“close to $900 million”“come close”“more closely”等与闭门会议无关的用法。

5 结果分析

5.1 TF-IDF 的余弦归一化偏向极短文档

TF-IDF 结果中短文档的集中，可以从公式 (2) 直接推出。对单词查询而言，查询向量归一化后权重为 1，文档得分退化为 $(1+\log_{10}\mathrm{tf}_{u,d})/\|d\|$。分子随词频只按对数增长，分母 $\|d\|$ 却大致随文档中不同词项的数量按平方根增长。以 Q1 为例，冰球预告只有 13 个词项，$\|d\| = 3.52$，“hurricanes” 只出现一次，得分为 $1/3.52 = 0.2841$；BM25 排名第 4 的 VOA 飓风报道长 577 个词项，“hurricane” 出现 16 次，分子为 $1+\log_{10}16 = 2.20$，但 $\|d\| = 19.54$，得分只有 0.1128。文档中十几次的重复提及抵不过分母的增长，于是只要查询词在一句简短的串场词里出现一次，这篇文档就能排到完整报道之前。

TDT3 的构成把这种偏向放大到了极端。语料中有 11,954 个 CNN 片段，长度中位数只有 63 个词项，其中大量是 “下节预告” 式的一句话。它们为热门话题提供了一批 “全文不超过 30 个词项、却含有查询词” 的候选文档：含 `hurricane` 的有 40 篇，含 `nba` 的有 18 篇，含短语 “newt gingrich” 的有 10 篇。这样的候选数量已经足以填满 Top-10，TF-IDF 的 Top-10 因此几乎被 CNN 包揽：Q4 为 10 篇，Q5 为 9 篇。多词查询的情况更不利：余弦得分是各单元贡献之和，而十几个词的短文档通常只容得下一两个查询词，于是 Q2 至 Q5 的 TF-IDF Top-10 中，含有全部查询单元的文档只有 3、3、0、0 篇，BM25 则为 10、10、9、1 篇，Q5 的 1 篇已是语料所能提供的上限。

这些短文档的主题判断未必错误，问题在于信息量和对查询条件的覆盖。TF-IDF 对 Q5 返回的十篇简讯全部涉及 NBA 停摆，但它们都只有一两句话，也没有覆盖查询中的 “strike” 和 “closed-door” 两个限定条件。相比之下，BM25 返回的完整报道对用户更有用。

5.2 BM25 的长度归一化与词频饱和

BM25 能避免上述偏向，依靠的是两个与 TF-IDF 不同的机制。第一是词频饱和：公式 (1) 中的分式随 $\mathrm{tf}_{u,d}$ 增长并趋于 k_1+1，一个查询词的贡献只取决于它自身的词频和文档长度；而在 lnc 方案中，文档里其他所有词项都会通过 $\|d\|$ 稀释查询词的权重。第二是长度归一化：BM25 的长度惩罚是相对于 avgdl 的线性插值，强度由 b 调节，余弦归一化则把每篇文档硬性缩放到单位长度，没有调节的余地。

b 值实验表明，第二个机制是结果长度特征的主要来源。$b=0$ 时长文因词频高而占优，Top-10 平均长度在 440 到 1,193 个词项之间；$b=1$ 时长度惩罚完全按比例施加，Q1 和 Q4 的 Top-10 退化为约 40 个词项的短片段，表现出与 TF-IDF 相似的偏向；$b=0.75$ 在两者之间取得平衡，Top-10 平均长度贴近语料平均值。由此可见，BM25 与 TF-IDF 在本实验中的差别，主要来自对文档长度的处理方式，而不是对词项权重的定义。

单词查询 Q1 的得分分布则体现了词频饱和的效果。十篇结果的得分都在上限 10.06 的 86% 至 88% 之间，彼此只差 0.21。全语料中“hurricane”出现次数最多的是一篇 NYT 长文，出现 22 次，但它长达 2,330 个词项，BM25 得分只有 7.30，没有进入 Top-10：高词频带来的增益已经饱和，抵不过长度带来的惩罚。

5.3 词干还原的两面作用

词干还原对五条查询的影响方向并不一致。Q2 是收益最明显的例子：“george”与“georges”被归并为同一词项，原本只代表人名的查询因此覆盖了飓风 Georges。启用词干还原时，Top-10 的十篇文档都同时含有两个查询词，且都是飓风报道；不做词干还原时，Top-10 有一半是靠 NBC 记者 George Lewis 的署名才命中的。不过这一收益带有偶然性：Porter 算法删去的是英语复数词尾“s”，它恰好与这个法语名字的拼写重合，预处理本身并不知道“Georges”是一个飓风的名字。

同一机制也会造成错误匹配。Q3 中，“billed”被还原为 `bill`，使 CNN 的一篇谈判报道因为“being billed”进入了 BM25 的 Top-10；Q1 中，TF-IDF 排名第 1 的冰球队“Hurricanes”也是经词干还原才与“hurricane”匹配。词干还原与小写化叠加，抹去了“Bill/bill”“Hurricanes/hurricanes”之间原本可以靠大小写和单复数区分的专名信息。

总体而言，词干还原在本实验中的作用偏正面。它提高了五条查询的召回，Q5 的命中数几乎翻倍；它引入的错误匹配集中在专名与普通词同形的场合，而在多词查询中，这类错误对排序的影响有限，因为一篇文档要排进前列，还需要同时命中其他查询词。Q3 的“Bill Hazard”一文就是如此：它的“Bill”是记者署名，但文档同时含有 `clinton` 和 `israel`，主体也确实是克林顿出访前的以色列局势。

5.4 短语与连字符的设计取舍

对于“newt gingrich”这样高度专指的短语，软匹配不会让不含短语的文档挤进前列，这是 Q4 中软、硬两种匹配结果一致的原因。具体有两层：其一，含有短语的文档比只零散含有组成词的文档多出一份短语单元的得分，而这个单元的逆文档频率很高（406 篇文档，BM25 逆文档频率为 4.53）；其二，“newt”和“gingrich”在语料中几乎总是相邻出现，409 篇同时含有二者的文档中只有 3 篇不相邻，软匹配额外纳入的候选大多只含其中一个词或只含 `down`，无法与含完整短语的文档竞争。软匹配真正发挥作用的场合，是短语在语料中很少出现甚至不出现的时候：硬匹配此时会直接返回空结果，软匹配仍能按组成词给出排序。

连字符实验说明，查询单元的个数本身就是一种隐性权重。“短语加组成词”的规则适合引号短语，因为用户用引号明确表达了“这几个词作为整体很重要”；套用到连字符词上，一个普通复合词就会获得三倍权重，Q5 的 Top-10 因而被“closed-door talks”这类与 NBA 无关的报道占据。改为一个连字符词只构成一个短语单元后，各查

询词在单元数上重新平等，排序回到以最具区分度的 `nba` 为主导。

“只拆成独立组成词”的规则同样使各查询词的单元数平等，而且因为不要求词序，它找到了一篇改写为“closed the door”的相关报道，这是仅短语规则做不到的。但放弃词序的代价同样明显：在词干还原的作用下，`close` 会匹配“close to”“closely”等意义完全不同的用法，Top-10 中有四篇文档就是这样得分的。两种规则在 NBA 相关文档数上持平，区别在于各自的误差：仅短语规则会漏掉词序被改写的表达，仅组成词规则则会混入大量与“闭门”无关的匹配。连字符复合词在英语中通常作为固定搭配出现，我们因此选择以词序约束换取匹配精度，把仅短语规则作为默认设置，同时保留 `--hyphen=parts` 选项供需要更高召回时使用。

5.5 系统效率

索引构建约需 22 秒，查询延迟在 14 毫秒以内，这主要得益于两个设计。其一，倒排表采用紧凑整数数组，加载时反序列化的对象少，索引加载只需约半秒；其二，打分按单元顺序累加，每个单元只遍历一次自己的倒排表，代价与查询词的文档频率之和成正比。Q3 和 Q4 含有 `clinton`、`down` 这样文档频率在 4,000 以上的词，打分耗时因此比只含低频词的 Q1 高出一个数量级，但仍在 3 毫秒以内。短语求交需要先展开各词的位置表，开销集中在含短语的 Q4 和 Q5 上，在当前规模下同样可以忽略。占用时间最多的是摘要生成，它需要从磁盘读取十篇原文并逐窗口扫描。

6 结论

我们用 Python 从零实现了一个面向 TDT3 语料的命令行搜索引擎，包括依原论文实现的 Porter 词干算法、对数字、缩写和连字符作针对性处理的分析器、基于紧凑整数数组的位置倒排索引、支持引号短语软匹配的查询解析器、BM25 与 TF-IDF 两种手写打分函数，以及查询相关摘要生成。37,526 篇文档的索引在约 22 秒内建成，五条测试查询的延迟都在 14 毫秒以内，返回条数可以通过 `--hits` 选项配置。

在五条测试查询上，BM25 返回的 Top-10 在主题上都与查询相符，并在多词查询上覆盖了尽可能多的查询单元。TF-IDF 的 lnc.ltc 方案则表现出强烈的短文档偏向，Top-10 平均长度只有 18 至 64 个词项，多由一两句话的广播串场词组成。BM25 的参数实验表明，二者差异的根源在于长度归一化方式：余弦归一化让 TDT3 中数量庞大的 CNN 短片段获得了不成比例的优势，而 BM25 取 $b = 0.75$ 时结果长度贴近语料平均水平。

对照实验为各项预处理与查询设计提供了依据。词干还原提高了全部五条查询的召回，并借“georges”的归并显著改善了 Q2，代价是在 Q3 等查询上引入了专名与普通词混同的错误匹配。保留停用词使 Q4 中的 `down` 得以参与排序，而它的权重已由逆文档频率适度压低。引号短语的软匹配在 Q4 上与硬匹配结果一致，在短语稀少时则能避免返回空结果。对连字符词，查询单元的个数本身就是一种隐性权重：把一个连字

符词只当作一个短语单元，能在保留词序信息的同时使各查询词的权重保持均衡，Q5 的 Top-10 中 NBA 相关报道因此从 1 篇增加到 8 篇，代价是会漏掉 “closed the door” 这类改写了词序的表达。

A TF-IDF（lnc.ltc）的 Top-10 结果截图

图 9 至图 13 是以 `--model=tfidf` 执行五条测试查询的控制台输出，其余设置与正文中的 BM25 结果相同。

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 --model=tfidf hurricane
q1 = hurricane (took 0.003 seconds)
[tfidf, 672 matching documents]

01 [0.2841] CNN19981128_1600_1369
finally, one matinee on ice. the hurricanes in new york, facing the islanders

02 [0.2601] CNN19981026_1600_0952
An already powerful hurricane is getting even stronger. Hurricane Mitch is now a
category 5 hurricane, with winds of 170 miles an hour. It's moving toward
central America and could reach Honduras and Nicaragua

03 [0.2524] CNN19981107_1600_0953
And the threat of disease is growing in central America, after hurricane mitch
struck the area

04 [0.2469] ABC19981025_1830_1017
Hurricane Mitch is now described as a very dangerous storm, and it's getting
stronger. Eastern Cuba are under a hurricane warning. Mitch's winds are about
150 miles an hour

05 [0.2377] CNN19981108_1000_0502
The extended weather forecast is next. After that -- American help for Honduras
in the aftermath of hurricane mitch

06 [0.2364] CNN19981026_2130_0931
Top stories this evening -- hurricane Mitch is growing into one of the strongest
atlantic storms ever recorded with sustained winds reaching 180 miles an hour.
The national hurricane center says the category 5 hurricane is at ...

07 [0.2358] CNN19981026_1130_0952
And the most powerful hurricane of the season is drifting slowly across the
western Caribbean, in the direction of central America. Hurricane Mitch has top
sustained winds of about 155 miles an hour

08 [0.2357] CNN19981027_2130_0519
The extended weather forecast is next. And still ahead -- using technology to
map hurricane damage along our coastlines

09 [0.2345] CNN19981027_0130_0931
Checking the top stories -- the national hurricane center says hurricane mitch
is an extremely dangerous storm which is capable of causing catastrophic damage.
Thousands of people are being evacuated from their homes

10 [0.2326] ABC19981130_1830_1182
This is the end of the hurricane season, according to the calendar, anyway. It
was the deadliest in 200 years and one of the most enduring. There is still
hurricane Nicole, way out in the atlantic, the tenth hurricane ...
```

图 9: Q1 `hurricane` 的 TF-IDF Top-10 结果



图 10: Q2 `mitch george` 的 TF-IDF Top-10 结果

```
python3 src/search.py

TDT3 search engine, 37526 documents indexed.
usage: search [--hits=N] [--model=bm25|tfidf] <query>   (quit to exit)
# search --hits=10 --model=tfidf bill clinton israel
q1 = bill clinton israel (took 0.007 seconds)
[tfidf, 7474 matching documents]

01 [0.2441] NBC19981220_1830_1169
When we continue -- peace and politics in Israel

02 [0.2236] CNN19981214_1600_0928
Checking our top stories -- President Clinton is commending Palestinian leaders
for efforts toward peace with Israel. Mr. Clinton was on hand in Gaza to see the
Palestine national council vote to reaffirm its commitment to remove ...

03 [0.2218] VOA19981208_0500_0050
I'm Bill Hazard, VOA news. U.S. mediator Dennis Ross is beginning a mission to
the mideast aimed at diffusing tensions between Israel and the Palestinians
ahead of President Clinton's trip to the region. Israel has suspended ...

04 [0.2202] PRI19981214_2000_0055
...this is Pamela McCall in the BBC Newsroom in London. The Palestinian's most
senior ruling body has once again renounced clauses calling for the destruction
of Israel. The Palestine National Council took the vote in front of Bill Clinton
...

05 [0.2158] CNN19981212_1000_0944
President Clinton is on his way to the mideast. He left for Israel this morning
to help usher in the second stage of the troubled wye river peace accord

06 [0.2140] VOA19981113_0500_2039
The Clinton Administration says it is disappointed with Israel's decision to
advertise for bids to build Jewish housing in east Jerusalem. The Israeli action
came a day after Israel's cabinet ratified the Wye River Peace Agreement ...

07 [0.2054] APW19981213_0389
The speaker of the Iranian Parliament said Sunday that U.S. President Bill
Clinton's planned visit to Gaza was part of a ``dangerous conspiracy,'' the
official news agency reported. Clinton, who arrived in Israel late Saturday,
plans ...

08 [0.2038] CNN19981017_1300_0938
Checking the hour's top stories -- President Clinton is joining and the leaders
of Israel and the Palestinian authority at the mideast summit in rural Maryland
today. Sources say little progress has been made

09 [0.2009] VOA19981208_0500_1832
...at diffusing tensions between Israel and the Palestinians ahead of President
Clinton's trip to the region. Mr. Ross is to meet first with Prime Minister
Benjamin Netanyahu in Israel, then hold talks with Palestinian leader Yassar
Arafat. President Clinton ...

10 [0.2005] CNN19981003_1600_0942
President Clinton is urging Congress to come up with a comprehensive Bill aimed
at protecting farmers. He wants republicanning to provide more money than
they're offering
```

图 11: Q3 `bill clinton israel` 的 TF-IDF Top-10 结果



图 12: Q4 `"newt gingrich" down` 的 TF-IDF Top-10 结果

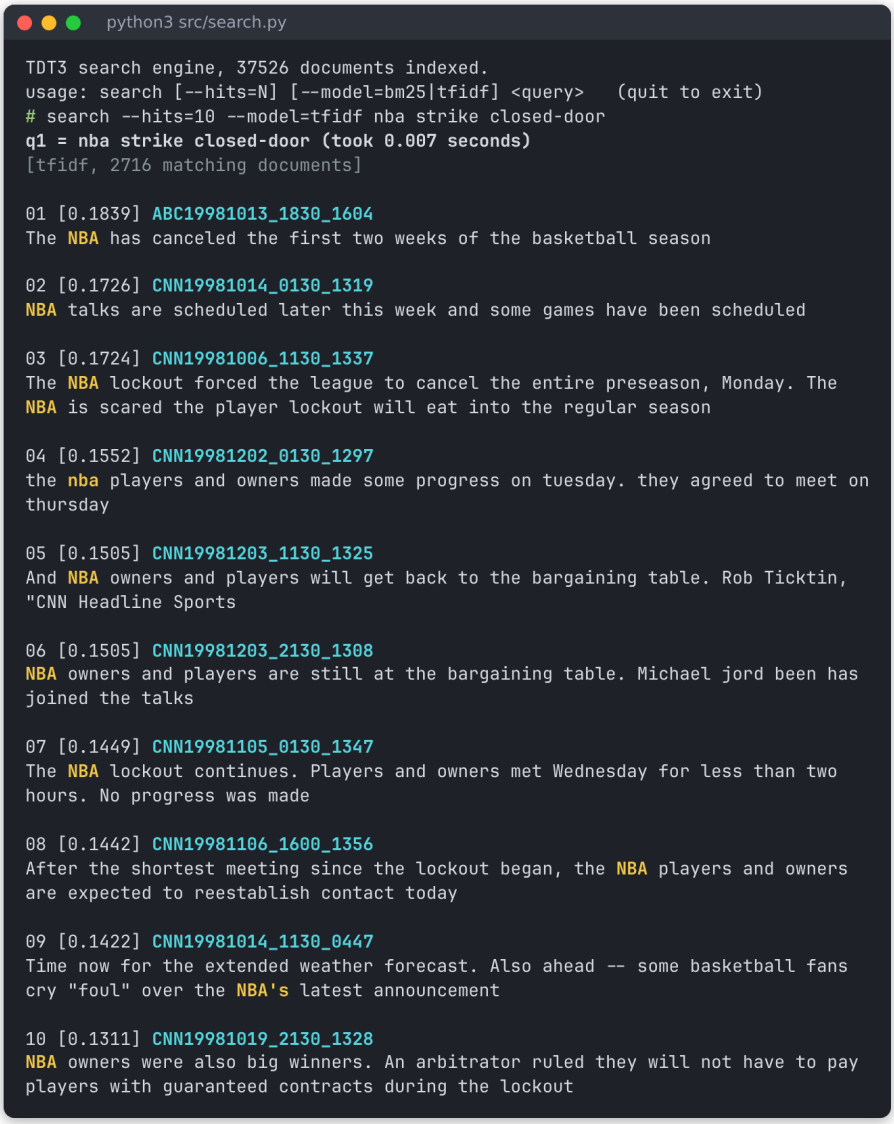


图 13: Q5 `nba strike closed-door` 的 TF-IDF Top-10 结果